# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines for feature engineering, processing approximately **1.5TB** of user interaction data weekly for downstream ML model training.
- Developed and deployed **FastAPI**-based microservices for real-time model inference, achieving p95 latencies under **150ms** for serving personalized recommendations to over **50 million** daily active users.
- Engineered **SQL** queries and data transformations on large datasets, optimizing analytical queries that reduced report generation times by an average of **30%** for business intelligence teams.
- Collaborated with product managers and stakeholders to gather requirements for new personalization features, translating business needs into technical specifications for ML model development.
- Analyzed user behavior patterns and experimental results from **A/B tests** to identify opportunities for improving product discovery and customer engagement, leading to a **4%** increase in conversion rates.
- Containerized ML services using **Docker** and deployed them to a Kubernetes cluster on **AWS EKS**, establishing CI/CD pipelines with **Jenkins** for automated testing and deployment.
- Investigated and resolved complex production issues in real-time personalization services, ensuring high availability and performance through proactive monitoring with **Splunk** and **Dynatrace**.

## SKILLS

**Languages** – Python, SQL, Java, English

**ML/AI** – Machine Learning, Personalization, Recommendation Systems, Ranking, Feature Engineering, Model Deployment, A/B Testing, scikit-learn, FastAPI, LightGBM

**Cloud/DevOps** – AWS, EKS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes, Jenkins, CI/CD, Git

**Data Technologies** – SQL, PostgreSQL, MySQL, Oracle, pandas, NumPy, Data Warehousing, ETL

**Frameworks/Tools** – Spring Boot, REST APIs, Flask, Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau

## PROJECTS

**Real-time Recommendation Service (HCL Technologies)**

- Developed a **Python**-based recommendation engine using **scikit-learn** and **pandas** to process user clickstream data, generating personalized product suggestions.
- Built a **FastAPI** service to serve model predictions in real-time, handling approximately **1,200 requests per second** under peak load.
- Integrated the service with an enterprise e-commerce platform, enabling dynamic content updates for logged-in users based on their interaction history.

**Customer Personalization Dashboard (HCL Technologies)**

- Created interactive **Power BI** dashboards to visualize user engagement metrics and the performance of personalization algorithms for internal stakeholders.
- Engineered **SQL** queries against PostgreSQL to extract and aggregate data for reporting, focusing on user segmentation and conversion funnel analysis.
- Designed data models in Power BI to support ad-hoc analysis and custom report generation, improving data accessibility for business analysts.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)